DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING (A)

Osmania University, Hyderabad-07

CLASS TEST-II

1005237290O8

SUBJECT:SIGNALS AND SYSTEM

Class: B.E. (AI&ML)

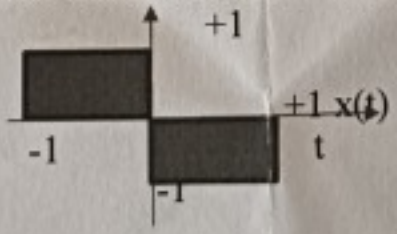
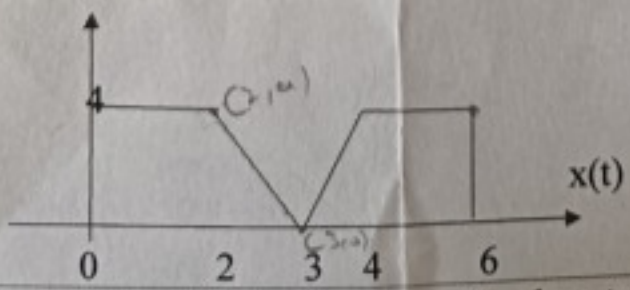
Semester: IV

Academic Year: 2024-25

Max Marks: 30

Duration:1 hr 30 min

Date:27-06-2025

	Answer All Questions			
	PART A (5 x 2 = 10 M)	Marks	BT	CO
1. a)	What are Dirichlet conditions of Fourier transforms	2	2	3
b)	Derive the relationship between laplace transform and z transform	2	2	5
c)	Find the Fourier transform of Gaussian pulse	2	2	3
d)	What are the advantages and limitations of z transform	2	1	5
e)	Determine Laplace transform of Impulse signal and sketch Region of Convergence.	2	2	3
	Answer any two PART B (2 x 10 = 20 M)			
2. a)	Find the the fourier transform of the wave form x(t) 	5	2	3
b)	Find the fourier transform of $e^{-2\|t\|}sgn(t)$	5	3	3
3. a)	Determine the laplace transform of the signal x(t) 	6	2	3
b)	State and prove final value theorem of Laplace transform	4	4	3

4. a)	Find the Z-transform of $x(n)=n\,2^n\sin(\pi n/2)\,u(n)$ and sketch ROC	5	3	5
b)	Consider a discrete time linear time invariant system described by the difference equation $y(n)-0.75y(n-1)+1/8\,y(n-2)=x(n)+1/3\,x(n-1)$ where $y(n)$ is the output and $x(n)$ is the input. Assuming that the system is relaxed initially obtain the unit sample response of the system	5	4	5